

Adverse Selection: a Strategic Foundation

Outline

- ① Adverse Selection and Market Breakdown
- ② Incentive Feasibility Under Adverse Selection

A General Model

We study the impact of private information on competitive allocations

Pricing in competitive markets under private information:

- General equilibrium theory. Agents only need to know prices and need not care whom they trade with.
- In some important situations, agents care about the identity of their counterparty

Common Values

Common Values: conditional on trade taking place, an agent's payoff directly depends on the characteristics, or information, of his trading partner

Examples Quality of a good (Akerlof 1970); productivity of a worker (Spence 1973); risk of an insuree (Rothschild-Stiglitz 1976); value of an asset (Leland-Pyle 1977)

Otherwise there are Private Values. Classical general equilibrium theory only allows for private values: agents do not trade with each other, but against their budget sets.

Common Values and Adverse Selection

Adverse selection arises when consumers and producers rank trades in opposite directions. For instance, in insurance markets:

1. At any given premium rate, riskier insurees demand higher coverage
2. But insurers are more reluctant to trade with them, as they are more likely to suffer a loss

Under adverse selection, competition can lead to no trade taking place in an equilibrium of the market for lemons (Akerlof 1970)

The Buyer

A buyer is privately informed of her type $i \in \{1, \dots, I\}$

$m_i \equiv$ probability of the buyer being of type i

Type i has utility $u_i(q, t)$ over quantities and transfers

$u_i(q, t)$ continuous and quasiconcave in (q, t)

Single Crossing

Types are ordered according to the (strict) single-crossing condition:

For all $i < j$, $q < q'$, t , and t' ,

$u_i(q, t) \leq (<) u_i(q', t')$ implies $u_j(q, t) \leq (<) u_j(q', t')$,

so type j is no less eager to increase her purchases than type i

Marginal Rates of Substitution

Without assuming differentiability, define

$\tau_i(q, t) \equiv$ slope of type i 's indifference curve at (q^+, t)

and make the following intuitive assumption:

Assumption 1 For all i and $q > 0$, $\tau_i(q, 0) \leq \tau_i(0, 0)$

Examples

Rothschild and Stiglitz (1976): The buyer faces a binary lottery $c_i W_B \oplus (1 - c_i) W_G$ over her wealth and has utility

$$U_i(Q, T) = c_i u(W_B - T + Q) + (1 - c_i) u(W_G - T).$$

$v_2 > v_1$ implies $MRS_2(Q, T) > MRS_1(Q, T)$ for all (Q, T) , so that the single-crossing property is satisfied.

Biais, Martimort, and Rochet (2000): The buyer is a privately-informed insider with CARA utility

$$U_i(Q, T) = \theta_i Q - \frac{\alpha \sigma^2}{2} Q^2,$$

for $\theta_2 > \theta_1$. The sellers are risk-neutral market makers, and v_i is the value of the asset conditional on the insider being of type i .

The Supply Side

The supply side is represented by a linear technology:

$c_i \equiv$ unit cost and $t - c_i q \equiv$ profit of serving type i ,

$\bar{c}_i \equiv \mathbf{E}[c_j | j \geq i] = \frac{\sum_{j \geq i} m_j c_j}{\sum_{j \geq i} m_j}$ upper-tail expectation of unit cost

Assume that c_i is nondecreasing in i , so that $\bar{c}_j \leq \bar{c}_i$ for all $j < i$

Inactive Markets

A market is inactive if, for whichever reason, no trading opportunity is available

Trade occurs through a contract $(q, t) \in \mathbb{R}_+^2$

Each type i 's reservation utility is $u_i(0, 0)$

When can an entrant profitably trade with the buyer?

Entry-Proofness in Inactive Markets: Necessary Conditions

Suppose an entrant wishes to attract type i of the buyer

He can do so with a contract (q, t) at price $p \equiv \frac{t}{q} < \tau_i(0, 0)$

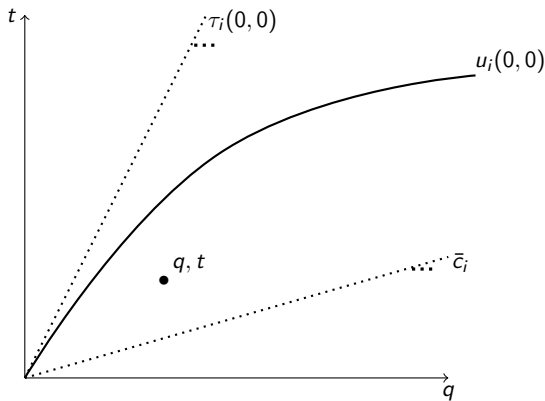
If $u_i(q, pq) > u_i(0, 0)$, then $u_j(q, pq) > u_j(0, 0)$ for all $j \geq i$

The entrant bears a unit cost $\bar{c}_i$ when trading with types $j \geq i$

The following is thus necessary for entry to be unprofitable:

Condition EP For each i , $\tau_i(0, 0) \leq \bar{c}_i$

$\tau_i(0, 0) > \bar{c}_i \longrightarrow (q, t)$ is profitable



Some types $j < i$ can be attracted as well, but for them $c_j \leq \bar{c}_i$

A General Result

More generally, the entrant may want to screen the buyer's type

Definition (Entry-Proofness):

*An inactive market is **entry-proof** if there is no menu of contracts that yields the entrant a strictly positive expected profit*

Theorem (Attar-Mariotti-Salanié 2021):

An inactive market is entry-proof if and only if Condition EP is satisfied

Sufficiency: Sketch of Proof

The buyer has a best response such that $q_{i-1} \leq q_i$ for each i

Summing by parts, the sellers' expected profit rewrites (Wilson 1993) as

$$\sum_i m_i(t_i - c_i q_i) = \sum_i \left(\sum_{j \geq i} m_j \right) [t_i - t_{i-1} - \bar{c}_i(q_i - q_{i-1})]$$

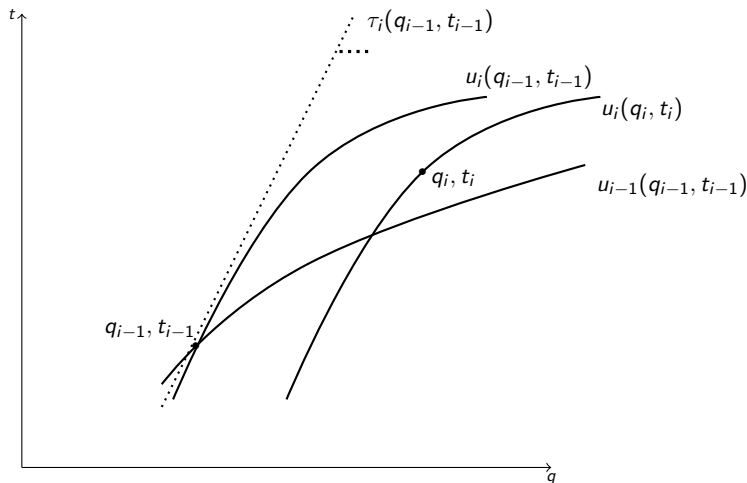
As type i is willing to trade the marginal quantity $q_i - q_{i-1}$,

$$t_i - t_{i-1} \leq \tau_i(q_{i-1}, t_{i-1})(q_i - q_{i-1}) \quad (*)$$

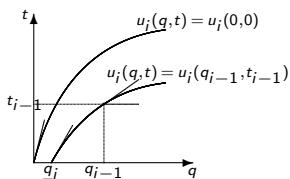
Thus profits are nonpositive if

$$\tau_i(q_{i-1}, t_{i-1}) \leq \bar{c}_i \text{ for all } i \quad (**)$$

Incentive Compatibility of type i



$$t_i - t_{i-1} \leq \tau_i(q_{i-1}, t_{i-1})(q_i - q_{i-1}) \quad (*)$$



$$\tau_i(q_{i-1}, t_{i-1}) \leq \tau_i(\underline{q}_i, 0) \text{ (by quasiconcavity);}$$

$$\tau_i(\underline{q}_i, 0) \leq \tau_i(0, 0) \text{ (by Assumption 1);}$$

$$\tau_i(0, 0) \leq \bar{c}_i \text{ (by Condition EP)}$$

Remarks on Condition EP

- Under adverse selection, the relevant measure of costs is $\bar{c}_i$
- One may have $\bar{c}_i \geq \tau_i(0,0) > c_i$ for all $i < I$
- Condition EP rules out gains from trade on *marginal* quantities
- Applies to a large class of economies: Quasilinear preferences, insurance economies with or without EU or RDEU, with smooth ambiguity aversion...

The Link with Wilson's (1993) Demand-Profile Approach

- Under quasilinear utility, a monopolist's expected profit writes as

$$\int_0^\infty \left(\sum_{j \geq i(q)} m_j \right) [U'_{i(q)}(q) - \bar{c}_{i(q)}] dq,$$

where $i(q)$ is the first type purchasing at least q .

- The solution to the monopolist's problem is given by

$$U'_i(q_i) = c_i + \frac{\sum_{j>i} m_j}{m_i} [U'_{i+1}(q_i) - U'_i(q_i)],$$

from which the $\bar{c}_i$ have disappeared.

The Second-Best Benchmark

When the planner perfectly controls trades, there is no loss of generality for him to offer an incentive-compatible, individually rational trading mechanism (Myerson 1979, 1982, Harris-Townsend 1981).

According to the Taxation Principle (Rochet 1985), such an allocation can simply be implemented by a tariff.

This allows one to characterize the second-best efficiency frontier in insurance economies (Prescott-Townsend 1984, Crocker-Snow 1985).

Second-Best: Feasibility

Consider a two-states $\{G, B\}$, two-types $\{1, 2\}$ insurance economy:

$$U_i(Q, T) = c_i u(W_B - T + Q) + (1 - c_i) u(W_G - T).$$

The array $\{(Q_1, T_1), (Q_2, T_2)\}$ is (IC) if:

$$U_i(Q_i, T_i) \geq U_i(Q_j, T_j) \quad \text{for all } i, j.$$

The trade (Q_i, T_i) is (IR) if:

$$U_i(Q_i, T_i) \geq U_i(0, 0) = c_i u(W_B) + (1 - c_i) u(W_G)$$

The array $\{(Q_1, T_1), (Q_2, T_2)\}$ is (BB) if:

$$\sum_i m_i (T_i - c_i Q_i) \geq 0.$$

Second-Best: Efficiency

Definition (Incentive Efficiency)

A second-best allocation is Pareto-efficient amongst those satisfying (IC), (IR) and (BB). Any such allocation $\{(Q_1^, T_1^*), (Q_2^*, T_2^*)\}$ is a solution of*

$$\mu U_1(Q_1, T_1) + [1 - \mu] U_2(Q_2, T_2)$$

for some $\mu \in [0, 1]$.

Second-Best: Efficiency (Cont'd)

Proposition (Crocker-Snow, 1985)

Any solution to the second-best problem involves a binding (BB). In addition, the following two cases obtain:

- ① *If $m_1 \geq \mu$ then $(IC)_1$ binds and type 1 receives full insurance.*
- ② *If $m_1 \leq \mu$ then $(IC)_2$ binds and type 2 receives full insurance.*

An Additional Constraint: Side Trading

Second-Best: individual types are unobservable, BUT individual trades can be perfectly monitored by the planning authority.

Few economic institutions have the ability to monitor all individual trades. We then need to include side trading in the theory of resource allocation under private information (Hammond (1979)).

⇒ Feasibility should incorporate the impossibility for the planner to control the trades between the buyer and the sellers.

Incentive Feasibility: Entry-Proofness

After the planner publicly offers a tariff specifying a transfer $T(Q)$ to be paid as a function of the quantity q demanded from him by the buyer, the sellers can offer additional contracts. Then, upon learning her type, the buyer chooses which quantity to demand from the planner and which contracts to trade with the sellers.

Instead of formally describing the strategic interaction between the sellers following the planner's tariff offer, we require that the tariff T resist entry by a seller offering contracts that the buyer can trade along with any quantity Q at price $T(Q)$.

The Market Tariff

Let the domain of T be compact and contain 0, with $T(0) = 0$.

Each type i picks a quantity $Q_i \in \arg \max \{u_i(Q, T(Q)) : Q\}$.

We say that the allocation $(Q_i, T(Q_i))_{i=1}^2$ is *implemented* by T .

Entry-Proofness

Definition (Entry-Proof Tariff)

*A tariff T is **entry-proof** if there is no menu of contracts that yields the entrant a strictly positive expected profit, given that the buyer is free to combine these contracts with the market tariff T*

Our goal is to characterize allocations that are

1. budget-feasible;
2. implementable by an entry-proof tariff.

The Jaynes–Hellwig–Glosten (JHG) Allocation

Each marginal quantity, or layer, is priced at the expected cost of serving the types who purchase it. The first layer Q_1^* is optimal for type 1 at unit price $\mathbf{E}[c_i] \equiv \bar{c}$,

$$Q_1^* \equiv \arg \max \{U_1(Q, \bar{c}Q) : Q \geq 0\},$$

$$T_1^* \equiv \bar{c}Q_1^*,$$

Then the second layer $Q_2^* - Q_1^*$ is optimal for type 2 at unit price c_2 , given that she already purchases the first layer Q_1^* ,

$$Q_2^* - Q_1^* \equiv \arg \max \{U_2(Q_1^* + Q, T_1^* + c_2Q) : Q \geq 0\},$$

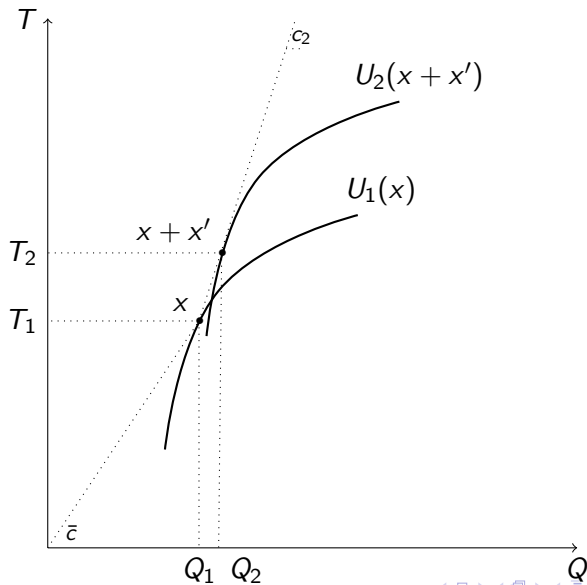
$$T_2^* - T_1^* \equiv c_2(Q_2^* - Q_1^*).$$

A Particular Implementation: Insurance

- 1) Both types buy a basic quantity Q_1 at price $\bar{c}$.
- 2) In addition, type 2 buys a quantity $Q_2 - Q_1$ at price c_2 .

In insurance, basic coverage at average premium complemented by optional coverage at a higher premium—notice the cross-subsidies!

Basic and Complementary Insurance



JHG: Insurance Economies

In the case of insurance, the JHG allocation stipulates that low-risk consumers obtain less than full coverage at the premium rate c , while high-risk consumers obtain full coverage by purchasing complementary coverage at the fair premium rate c_2 .

Because, for a high-risk consumer, the optimal complementary coverage at the fair premium rate v_2 is positive, her incentive-compatibility constraint is slack, $U_2(Q_2^*, T_2^*) > U_2(Q_1^*, T_1^*)$.

$\implies$ The JHG allocation is thus not second-best efficient.

Why Not Increase Coverage for Low-Risk Consumers?

Increasing the coverage sold to the low-risk consumer beyond Q_1^* at a unit price in $(c_1, MRS_1(Q_1^*, T_1^*))$ is not an option for the planner because it makes it feasible for an entrant to profitably attract the high-risk consumer with complementary coverage at a premium rate slightly below the fair premium rate c_2 .

Because the high-risk consumer can combine such coverage with the coverage intended by the planner for the low-risk consumer, an entrant can exploit the trades offered by the planner to make a profit, leading to a deficit for the planner.

In fact, $U_2(Q_2^*, T_2^*) = \max\{U_2(Q_1^* + Q, T_1^* + c_2 Q) : Q \geq 0\}$, which is a multiple-contracting analogue of a binding incentive-compatibility constraint for type 2. Does this mean that the JHG allocation is “third-best efficient”? Yes, and much more!

The Main Characterization Result

Theorem (Attar-Mariotti-Salanié, 2020)

The JHG allocation is the only budget-balanced allocation that is implementable by an entry-proof tariff.

Proof: Uniqueness

Any such allocation must satisfy

$$U_1(Q_1, T(Q_1)) \geq \max\{U_1(Q, \bar{c}Q) : Q \geq 0\},$$

$$U_2(Q_2, T(Q_2)) \geq \max\{U_2(Q_1 + Q, T(Q_1) + c_2Q) : Q \geq 0\},$$

for, otherwise, an entrant can offer profitable contracts at unit prices slightly above $\bar{c}$ or c_2 , respectively. But then

$$T(Q_1) \leq \bar{c}Q_1,$$

$$T(Q_2) \leq T(Q_1) + c_2(Q_2 - Q_1),$$

so that the budget-balance constraint

$$T(Q_1) - \bar{c}Q_1 + m_2[T(Q_2) - T(Q_1) - c_2(Q_2 - Q_1)] \geq 0$$

is satisfied if and only if all these inequalities are in fact equalities, which then in effect define the JHG allocation.

Proof: Existence (1)

The piecewise-linear convex tariff

$$t^G(q) \equiv 1_{\{q \leq Q_1^*\}} \bar{c}q + 1_{\{q > Q_1^*\}} [\bar{c}Q_1^* + c_2(q - Q_1^*)]$$

implements the JHG allocation. Denote by (q_1, t_1) and (q_2, t_2) the trades type 1 and type 2 make with an entrant.

Because t^G is convex, the buyer has a best response such that $q_2 \geq q_1$. As t^G allows type 1 to buy her optimal quantity Q_1^* at unit price $\bar{c}$, one must first have

$$t_1 \leq \bar{c}q_1 \quad (a)$$

Proof: Existence (2)

Let $(\hat{Q}_2, \hat{T}_2)$ be the post-entry aggregate trade of type 2.

Because $\hat{Q}_2 \geq q_2 \geq q_1$, type 2 can trade (q_1, t_1) with the entrant and demand $\hat{Q}_2 - q_1$ from the planner, paying $t_1 + t^G(\hat{Q}_2 - q_1)$ for $\hat{Q}_2$.

She trades $(\hat{Q}_2, \hat{T}_2) \implies \hat{T}_2 \leq t_1 + t^G(\hat{Q}_2 - q_1)$

Recall that $\hat{T}_2 = t_2 + t^G(\hat{Q}_2 - q_2)$

$$\implies t_2 - t_1 \leq t^G(\hat{Q}_2 - q_1) - t^G(\hat{Q}_2 - q_2) \quad (b)$$

t^G is convex with slope at most c_2 and $\hat{Q}_2 - q_1 \geq \hat{Q}_2 - q_2$ as $q_2 \geq q_1$,

$$\implies t^G(\hat{Q}_2 - q_1) - t^G(\hat{Q}_2 - q_2) \leq c_2(q_2 - q_1) \quad (c)$$

(a)–(c) imply that the expected profit of the entrant is at most zero. ■

Remark I : Characterization

The JHG allocation is the unique entry-proof allocation even when the entrant is restricted to offer a single contract, whereas the tariff t^G resists entry even when the entrant can offer an arbitrary tariff.

This characterization result is very general, and does not even rely on an expected-utility representation for the buyer's preferences, unlike the characterization of the second-best efficiency frontier provided by Prescott and Townsend (1984) and Crocker and Snow (1985).

Remark II: Markets as a Constraint

Entry-proofness characterizes a unique allocation.

This shows that the planner is severely constrained by his inability to control the buyer's trades with a potential entrant. The threat of such trades effectively deprives him of any possibility to transfer utility between the two types.

Existence is not an issue, unlike in Rothschild and Stiglitz (1976).

The difference is that an entrant in the exclusive case can fully control the buyer's trades, while, in this setting, any contract he may offer can be combined by the buyer with the planner's and the other sellers' offers.

Whereas this enlarges the set of contracts an entrant can offer to attract the buyer, this also gives the planner more instruments to deter entry.

Remark III: Allocations, Tariffs, and Latent Contracts

When the planner perfectly controls trades, there is no need, according to the Revelation Principle, to distinguish tariffs from allocations.

However, when sellers can propose additional contracts on top of the planner's tariff, the planner may have to issue additional, latent contracts, which are not traded by the buyer but are only meant to block entry.

Whereas such latent contracts may be necessary, the planner must make sure that, by merely offering them, he does not create other profitable entry opportunities. The constructed tariff t^G strikes a balance between these two constraints.